

AR RAQQA, Syrian Arab Republic

WMO: 400390

Lat: 35.933N

Lon: 39.017E

Elev: 807

StdP: 14.27

Time Zone: 2.00 (SYR)

Period: 82-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	30.8	15.6	12.5	37.5	19.1	14.8	39.8	18.5	51.1	13.9	50.9	0.3	290	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.8	105.7	72.5	102.7	71.3	100.1	70.4	75.9	99.0	74.4	96.7	73.0	94.4	3.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.9	109.5	88.3	67.1	102.7	87.4	65.5	96.9	87.0	40.0	99.2	38.5	96.8	37.3	94.2	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.5	11.0	9.3	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	23.0	78.9	3.0	3.8	20.9	81.6	19.1	83.8	17.4	85.9	15.2	88.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.9	45.2	49.2	56.0	65.3	74.1	83.6	87.8	87.2	79.7	70.0	56.4	47.3
	DBStd	16.07	5.45	5.65	6.57	6.34	5.30	4.23	3.98	4.10	4.69	5.33	6.41	5.43
	HDD50	400	166	75	21	1	0	0	0	0	0	0	20	117
	HDD65	2244	615	442	287	69	3	0	0	0	0	19	261	548
	CDD50	6576	16	53	207	461	749	1009	1171	1152	891	621	212	34
	CDD65	2944	0	0	8	79	287	559	706	687	441	174	3	0
	CDH74	38155	0	0	106	870	3139	7567	10330	9637	4940	1523	43	0
(13)	CDH80	21423	0	0	17	300	1408	4454	6426	5838	2481	498	1	0
(14)	Wind	WSAvg	2.6	1.6	2.4	2.6	2.7	2.9	4.1	5.0	3.9	2.4	1.5	1.1
(15) Precipitation	PrecAvg	7.0	1.5	1.1	1.1	0.8	0.5	0.0	0.0	0.0	0.0	0.3	0.8	1.0
	PrecMax	12.3	4.9	3.2	4.6	2.5	4.0	0.2	0.1	0.0	0.3	1.9	3.1	4.0
	PrecMin	2.2	0.2	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2
	PrecStd	2.8	1.0	0.7	1.1	0.7	0.8	0.0	0.0	0.0	0.1	0.4	0.8	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	68.4	82.5	94.3	97.7	108.0	109.5	109.5	99.5	91.4	78.6	67.0
		MCWB	53.3	53.2	60.8	64.7	66.7	71.3	73.3	75.0	72.8	66.3	58.9	51.6
	2%	DB	59.4	65.2	76.7	87.5	94.1	102.8	106.0	105.3	96.5	88.2	74.8	61.7
		MCWB	50.6	51.8	57.0	63.0	65.3	69.2	71.7	74.3	71.5	65.8	58.0	51.1
	5%	DB	56.7	62.4	73.0	83.0	90.8	99.8	103.1	102.2	94.3	85.5	72.0	58.9
		MCWB	49.1	50.9	56.0	61.3	64.2	67.5	70.7	73.3	70.4	65.3	57.1	50.0
	10%	DB	54.0	59.2	69.3	78.9	87.8	97.1	100.3	99.2	91.9	82.5	68.7	56.2
		MCWB	47.5	49.3	54.3	59.4	63.5	66.9	70.1	72.1	69.3	64.0	55.8	48.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	55.9	62.5	67.5	69.3	75.1	77.9	78.4	75.7	70.0	61.9	55.9
		MCDB	62.2	65.4	77.1	87.9	91.8	102.5	100.2	101.2	94.6	86.3	73.3	61.2
	2%	WB	52.3	53.2	59.7	64.5	67.3	71.7	75.0	76.4	73.4	67.8	59.9	53.4
		MCDB	57.5	62.5	73.0	83.2	89.7	96.3	98.8	99.7	92.8	84.8	71.2	59.0
	5%	WB	50.2	51.8	57.3	62.5	65.6	70.1	73.5	75.0	72.1	66.2	58.4	51.7
		MCDB	55.3	59.7	69.3	80.4	86.8	94.1	96.7	97.9	90.7	82.5	69.4	56.7
	10%	WB	48.4	50.5	55.2	60.5	64.3	68.7	72.1	73.7	70.8	64.7	56.7	50.0
		MCDB	52.7	57.8	66.9	76.2	84.2	92.1	95.4	95.3	88.8	80.2	66.7	55.0
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	17.8	21.5	22.6	24.5	25.6	24.8	24.5	24.0	24.2	22.3	19.1
		MCDBR	19.2	22.2	27.3	29.3	28.5	29.4	27.8	27.4	26.1	28.5	26.7	21.6
		MCWBR	11.0	11.1	12.8	12.2	10.5	10.2	9.9	11.5	11.8	12.9	13.4	12.5
	5% WB	MCDBR	15.9	19.0	23.8	26.4	26.8	27.1	25.4	25.9	24.6	25.8	22.7	17.8
		MCWBR	9.9	10.5	11.9	11.8	10.9	10.2	10.0	11.8	12.2	12.4	11.7	11.2
(40) Clear-Sky Solar Irradiance	taub		0.378	0.407	0.433	0.501	0.494	0.439	0.432	0.439	0.416	0.454	0.392	0.378
	taud		2.156	2.070	2.013	1.855	1.906	2.074	2.110	2.085	2.136	2.004	2.137	2.155
	Ebn at Noon		250	258	263	251	255	269	269	265	265	238	242	240
	Edh at Noon		36	45	52	64	62	52	50	50	45	47	37	35
(44) All-Sky Solar Radiation	RadAvg		794	1072	1524	1888	2250	2574	2496	2225	1852	1337	963	724
	RadStd		64	100	82	108	123	81	51	66	71	75	78	75

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	-3.19	N/A	-1.60	N/A	N/A	N/A	-97	N/A	N/A

Nomenclature: See separate page